WP 314

Predicting Immune Cell Admixture in Bulk Proteomics Data Using ProteoMixture

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Background

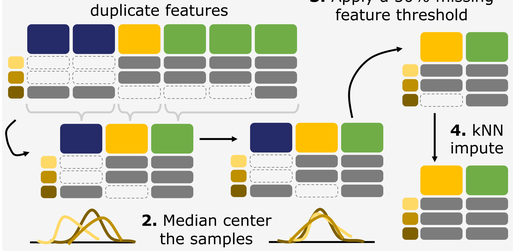
- ProteoMixture[1] accurately estimates relative tumor, stromal, and immune cell populations from bulk tissue proteomic data.
- xCell[3] infers immune cell subtype enrichment from transcriptomics.
- Transcript levels poorly reflect protein abundance, underscoring the need for proteomics-based tools.

Objective

Discover and validate protein features predicting immune types in proteomics data.

Methods

CPTAC[2] Data Preparation *For Each Tumor Type*



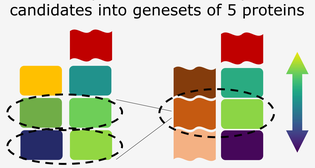
Creating ProteoMixture Genesets

For Each Geneset

1. Combine all 10 tumor type matrices, dropping any features that are not found in all tumor types

2. Get xCell ssGSEA scores for transcript

3. Take highly correlated proteins (abundance to ssGSEA score) as protein genesets, remove least correlated protein and distill optimized candidates into genesets of 5 proteins



4. Choose highest correlation for ProteoMixture geneset

Results

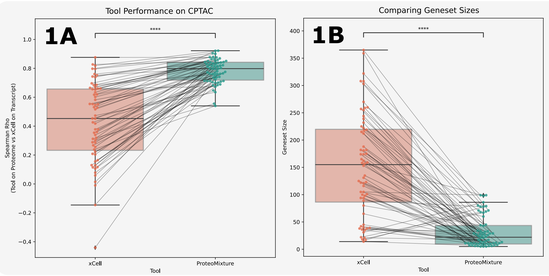


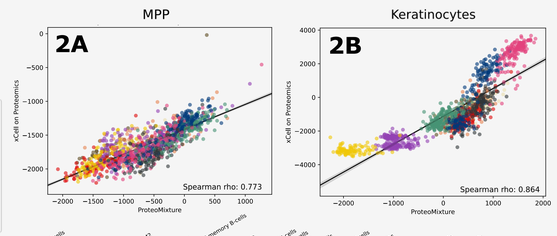
Figure 1A. xCell and ProteoMixture geneset ssGSEA scores on CPTAC tumor proteomes correlated to equivalent xCell scores on transcriptomes across all cancer types.

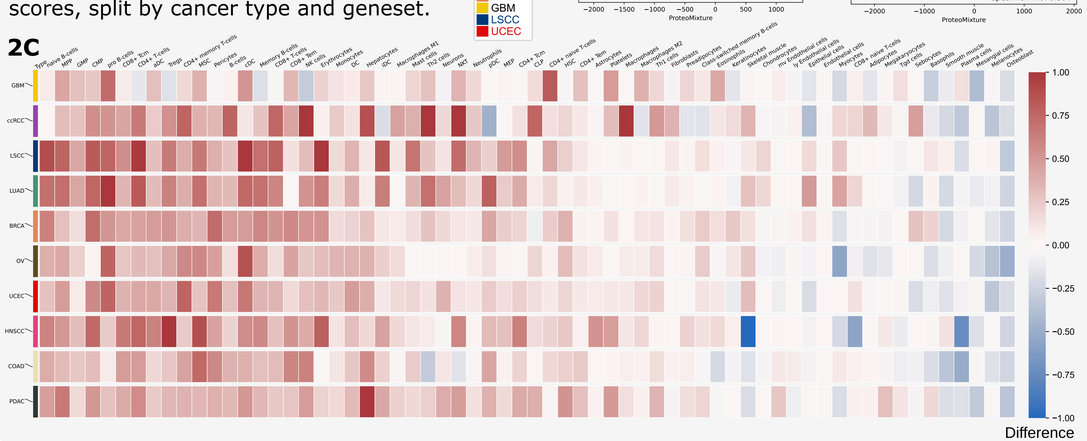
1B. xCell (combined) and ProteoMixture geneset sizes.

Annotations represent unadjusted two-tailed paired Wilcoxon test.

Figure 2A, 2B. CPTAC ProteoMixture versus xCell ssGSEA scores for Multipotent Progenitors (MPP) (**2A**) and Keratinocytes (**2B**) across all cancer types.

2C. ProteoMixture protein scores correlated to xCell transcript scores subtracting xCell protein scores correlated to xCell transcript scores, split by cancer type and geneset.





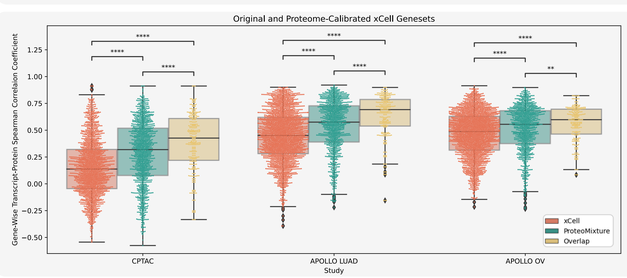
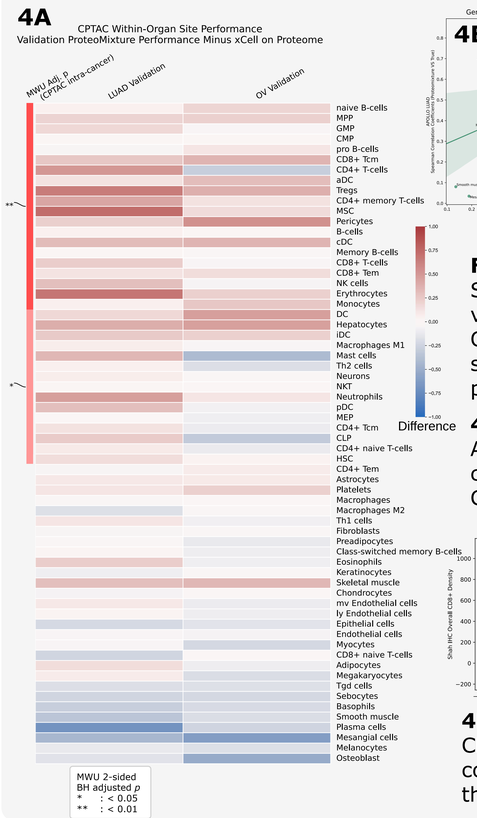


Figure 3. Gene-wise RNA & protein correlations (GRPCs) for each gene in the xCell genesets, ProteoMixture immune subtype genesets, and genes which exist in both.

Annotations represent unadjusted two-tailed Mann-Whitney U test results.

MWU 2-sided *p*
* : < 0.05
** : < 0.01
*** : < 0.001
****: < 0.0001



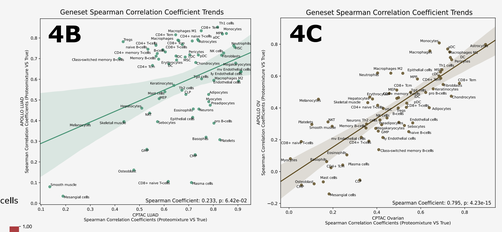
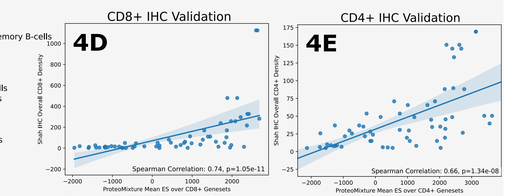


Figure 4A. ProteoMixture/xCell geneset Spearman correlation differences within validation APOLLO LUAD[4] and APOLLO OV[5] samples. ProteoMixture genesets are sorted by overall CPTAC intra-cancer performance when compared to xCell.

4B, 4C. APOLLO LUAD[4] (**4B**) and APOLLO OV[5] (**4C**) ProteoMixture/xCell correlations compared to the respective CPTAC cancer type.



4D, 4E. Zhang[6] ProteoMixture average CD8+ (**4D**) and CD4+ (**4E**) scores correlated to cell density determined by the respective IHC stain.

Conclusions

ProteoMixture provides immune cell subtype enrichment scores from protein abundance.

- ProteoMixture's proteome-optimized features replicate xCell's transcript-based scores, allowing for inter and intra-cancer proteomic comparisons. (**Figures 1, 2**)
- ProteoMixture's optimized features discovered in CPTAC[2] were validated in APOLLO LUAD[4] and APOLLO OV[5] (**4A, 4B, and 4C**), and partially with IHC. (**4D, 4E**)

ProteoMixture Online Tool

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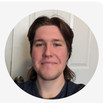

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4 – Soltis et al. "Proteogenomic analysis of lung adenocarcinoma reveals tumor heterogeneity, survival determinants, and therapeutically relevant pathways." *Cell Reports Medicine*, 2022, https://doi.org/10.1016/j.xcrm.2022.100819
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